

EE 210L: Digital Circuits and Logic Design Lab

Lab 6: Latches and Flip-Flops

Name: _____ Date: _____

Partner(s): _____ Section: _____

The 74LS73 uses V_{CC} on pin 4 and GND on pin 11. Confirm your wiring before applying power.

Part 5.1: NAND SR Latch

$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	Behavior (set / reset / hold / forbidden)
1	1			
0	1			
1	1			
1	0			
1	1			
0	0			

5.1.1 Forbidden State: Release Test

Take both inputs LOW, then release both at once. Record where the latch lands. Repeat five times.

Trial	1	2	3	4	5
Q after release					

Was the result the same every time? Explain what determines the outcome.

Part 5.2: 74LS279 Quad SR Latch

$\bar{S}$	$\bar{R}$	Measured V at Q (V)	Q	Behavior
1	1			
0	1			
1	0			
0	0			

Part 5.3: D Flip-Flop (74LS74)

5.3.1 Clocked Operation

D	Q after several clock edges	$\bar{Q}$
1		
0		

5.3.2 Transparency Check

Clock held at	Does Q follow changes on D?
HIGH (static)	
LOW (static)	

What does this prove about the difference between a latch and an edge-triggered flip-flop?

5.3.3 Asynchronous Clear

With the clock running and D held HIGH, pulsing $\bar{CLR}$ LOW caused Q to:

Did it wait for a clock edge?

Part 5.4: Propagation Delay

Paste your oscilloscope capture showing the clock edge and the Q transition with cursors placed.

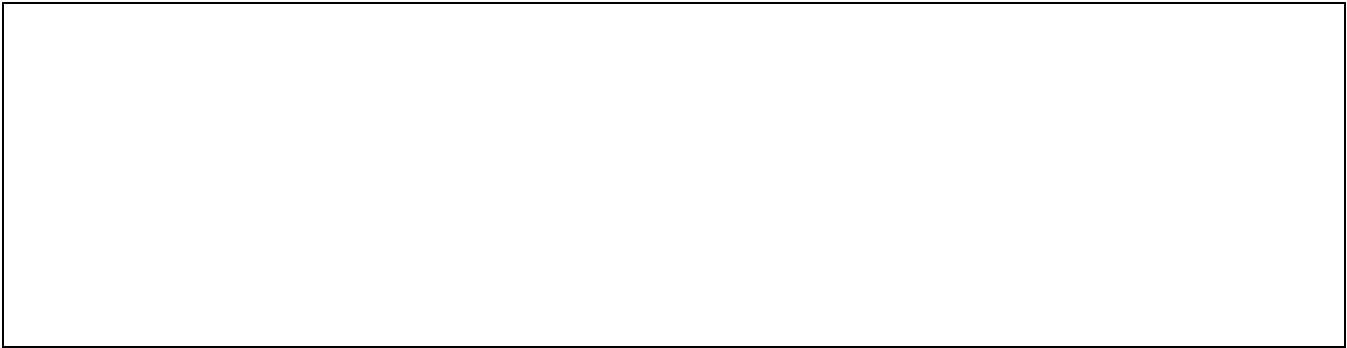


Figure 1: Clock-to-Q propagation delay measurement.

Measured t_{PLH} or t_{PHL} (clock edge to 50% of Q): _____ ns

Datasheet value for the 74LS74: _____ ns

If they differ, suggest a reason.

Part 5.5: JK Flip-Flop (74LS73)

J	K	Observed Q behavior	Expected behavior	Match?
0	0			
0	1			
1	0			
1	1			

Part 5.6: Toggle Mode and Frequency Division

Paste your capture showing the clock and Q together.

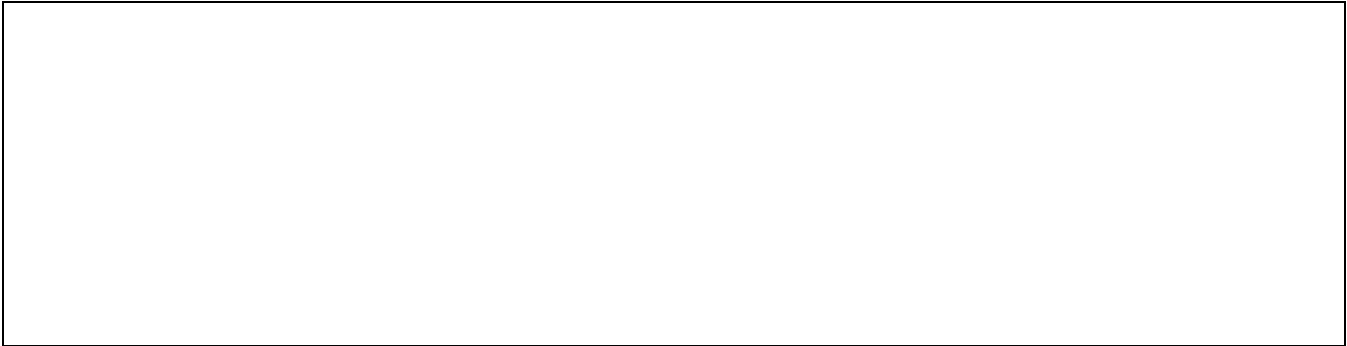


Figure 2: Clock and Q of a toggle flip-flop.

Signal	Measured frequency	Ratio to clock
Clock		1
Q of first flip-flop		
Q of second flip-flop		

A chain of toggle flip-flops is counting something. What?

Discussion and Conclusions

Address at least: why the SR forbidden state is genuinely unpredictable rather than merely undefined on paper, why synchronous systems are built from edge-triggered flip-flops rather than latches, and what setup and hold

time violations would do to your circuit.
